

als. Bharti Teletech has already started recruiting in this field.

Find your role

SMT is broadly dedicated to electronics and telecom professionals. Most companies look for degree or diploma holders with some hands-on experience in electronics. The scope for post-graduates or doctorate degree holders is rather limited. More precisely, in-hand experience may be more advantageous than higher educational degrees. You will find no threat from your contemporary fellows with IT or computer science background. But, at the same time, you should be prepared for a typical blue-collar job.

As an SMT engineer, you will work either at the manufacturing plant handling the process line or in the service sector dealing with maintenance and troubleshooting of SMT machines. Often, it involves quality control and shop-floor management. The designation and responsibilities may change with time but it is almost impossible to get rid of hardware if you decide to build a foothold in this sector.

What you can get?

The SMT sector offers a decent starting salary in the range of Rs 8000 to Rs 10,000 per month (excluding allowances) for diploma holders and Rs 15,000 to Rs 20,000 for degree holders.

Though the SMT sector is not as high-paying as the embedded sector, it is an ideal field for steady job seekers with average competency. According to Pradyunma Salaria, deputy manager-SMT, American Tec, a diploma holder from even a not-so-much-reputed institute may earn around Rs 500,000 per annum with 4-5 years of experience in this field. The package could be Rs 700,000-Rs 800,000 for degree holders with similar experience. But to climb the ladder, you need to keep a soft corner for your soft skills. Because, with the progress of your career, you will be expected to lead a production team towards productivity improvement by designing projects

One may continue career either in SMT services or in SMT manufacturing. But the growth prospects are higher in the latter case.



—Vishal Dahiya, Factory Automation, Panasonic

and new processes, in spite of the technical nature of the SMT jobs.

The workshop is revealed

It is quite possible that you never came across the term 'SMT' in your textbook. Don't get disappointed. Two top executives of American Tec, Avishek Tyagi, regional manager-sales and marketing, and Pradyunma Salaria, deputy manager-SMT, reveal that the basic knowhow of electronics as well as other allied fields like electrical engineering and mechanical engineering is quite important. However, they specifically mention that it is no longer enough to know the fundamentals. The professionals are operating in a very cut-throat environment, so they should know not only how to survive but also compete.

N. Chandramohan, country head-SMT division, Juki India, opines that "proper training along with professional degree would be the turnkey for a fresher. This training should be totally industry-focused." He feels that an effective collaboration between the industry and academia may be fruitful for both the parties. The existing industry-institute partnership models of some reputed foreign universities are ideal to follow.

From chip to ship

Undoubtedly, this training should be balanced enough to deal with both theoretical and practical aspects of SMT. Tyagi suggests that "a holistic training for the entry-level professionals in the field of SMT should cover the basic theory and application of SMT operation, process flow of SMT,

soldering technique and solder pastes, basic principles of machineries used for SMT, software and hardware related to the SMT process, maintenance and quality control procedures and protocols in the SMT process and also storage, shipping and handling of SMT products."

Besides the basics and applications of SMT, you need to know the basics of electronic circuit designing and also production technology.

Keep in mind that SMT is a whole new package responsible for improved integration of the circuit. That's why the reliability of both chip-to-chip and chip-to-device interconnections is important. Basically, SMT reduces the package size resulting in greater functionality in the same board area. Mobile and handheld electronic items such as video cameras and cellphones are examples of low weight, high performance and also reduction in noise (primarily due to smaller electrical paths than leaded components) achieved with SMT. SMT is particularly useful for RF and microwave circuits where low noise contribution is mandatory. Moreover, it results in a higher operating speed due to shorter interconnect distances.

At first, today's device requirements appear to oppose each other. SMT solves this problem in a different way. New types of components, new mounting and testing techniques and a new set of design guidelines are dedicated for this purpose.

SMCs: The building blocks

You need to have a clear picture of surface-mount components (SMCs) as these